Machine Learning Lab

Experiment -06

Regression and Metrics

Name: Malli Gowtham Sai

Reg Number: 23BCE20092

Course Name: Introduction to Machine Learning

Course Code: CSE3008

Slot: L26+L27

Faculty: Siddique Ibrahim Peer Mohamed

Dataset

Title: Fuel Production vs Consumption

Type: Regression Dataset

Format: csv file

Source:https://www.kaggle.com/datasets/shawkatsujon/worldwide-fuel-production-and-consumption

Year	Entity	Gas produ	Gas consu	Coal produ	Coal consu	Oil product	Oil consun	Gas produ	Gas consu	Coal produ	Coal consu	Oil product	Oil consun	Population
1980	Afghanista	1.7E+09	56640000	119000	119000	0	406500	127.2	4.241	0.00891	0.00891	0	0.03043	13360000
1981	Afghanista	2.24E+09	84960000	125000	125000	0	464600	169.9	6.45	0.00949	0.00949	0	0.03527	13170000
1982	Afghanista	2.29E+09	1.42E+08	145000	145000	0	452900	178.1	10.99	0.01126	0.01126	0	0.03516	12880000
1983	Afghanista	2.41E+09	1.42E+08	145000	145000	0	638800	192	11.29	0.01157	0.01157	0	0.05095	12540000
1984	Afghanista	2.41E+09	1.42E+08	148000	148000	0	638800	197.2	11.6	0.01213	0.01213	0	0.05234	12200000
1985	Afghanista	2.97E+09	0	151000	151000	0	754900	249.1	0	0.01265	0.01265	0	0.06323	11940000
1986	Afghanista	2.97E+09	0	160000	160000	0	789600	253.4	0	0.01363	0.01363	0	0.06728	11740000
1987	Afghanista	2.8E+09	6.23E+08	167000	167000	0	805300	241.6	53.69	0.01439	0.01439	0	0.0694	11600000
1988	Afghanista	3E+09	1.98E+09	138000	138000	0	765700	258.4	170.6	0.01188	0.01188	0	0.06591	11620000
1989	Afghanista	2.95E+09	1.93E+09	127000	127000	0	776600	248.2	162.3	0.0107	0.0107	0	0.06543	11870000
1990	Afghanista	2.95E+09	1.9E+09	105000	105000	0	788200	237.3	152.9	0.008459	0.008459	0	0.0635	12410000
1991	Afghanista	3.12E+08	3.12E+08	94000	94000	0	740700	23.42	23.42	0.007068	0.007068	0	0.05569	13300000
1992	Afghanista	3E+08	3E+08	8000	8000	0	414400	20.71	20.71	0.000552	0.000552	0	0.02861	14490000
1993	Afghanista	3E+08	3E+08	7000	7000	0	414400	18.97	18.97	0.000443	0.000443	0	0.0262	15820000
1994	Afghanista	3E+08	3E+08	6000	6000	0	385400	17.56	17.56	0.000351	0.000351	0	0.02257	17080000
1995	Afghanista	2E+08	2E+08	5000	5000	0	373800	11.04	11.04	0.000276	0.000276	0	0.02064	18110000
1996	Afghanista	2.3E+08	2.3E+08	3000	3000	0	343800	12.2	12.2	0.000159	0.000159	0	0.01824	18850000

Steps:

1. Importing packages

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor

from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
```

2. Dataset Cleaning and display

```python
# Check missing values
print("\nMissing Values:")
print(dataset_df.isnull().sum())

dataset_df.fillna(dataset_df.median(numeric_only=True), inplace=True)


for col in dataset_df.select_dtypes(include='object').columns:
    dataset_df[col] = dataset_df[col].fillna(dataset_df[col].mode()[0])

dataset_df = pd.get_dummies(dataset_df, drop_first=True)

print("\nCleaned Dataset:")
print(dataset_df.head())
```

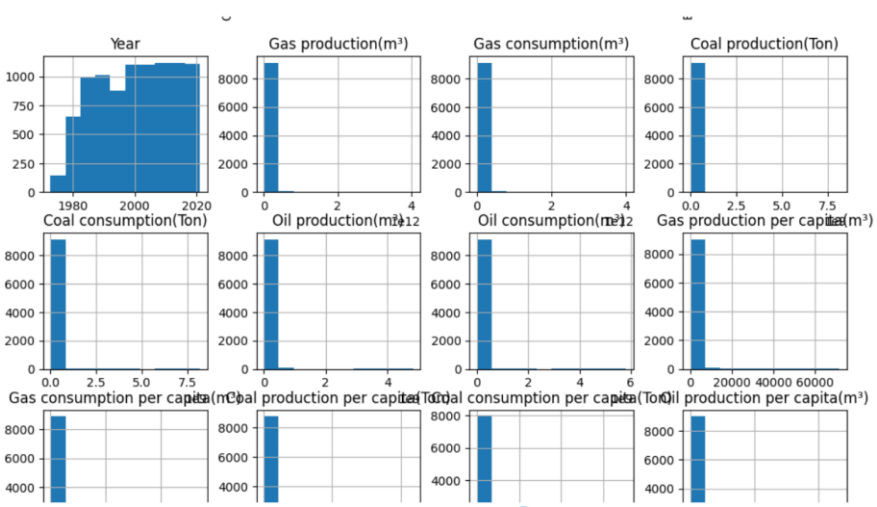
```
Coal production per capita(Ton)     1480
Coal consumption per capita(Ton)    1480
Oil production per capita(m³)       1417
Oil consumption per capita(m³)      1636
Population                          1105
dtype: int64

Cleaned Dataset:
   Year  Gas production(m³)  Gas consumption(m³)  Coal production(Ton)  \
0  1980        1.699000e+09           56640000.0              119000.0
1  1981        2.237000e+09           84960000.0              125000.0
2  1982        2.294000e+09          141600000.0              145000.0
3  1983        2.407000e+09          141600000.0              145000.0
4  1984        2.407000e+09          141600000.0              148000.0
```

3.Exploratory Data Analysis(EDA)

```python
plt.figure(figsize=(10,8))
sns.heatmap(dataset_df.corr(), annot=True, cmap="coolwarm")
plt.title("Feature Correlation Heatmap")
plt.show()

# Distribution Plot
dataset_df.hist(figsize=(12,10))
plt.show()
```



4.Feature Selection

Feature Selection

```
target = "Gas consumption(m³)"

X = dataset_df.drop(target, axis=1)
y = dataset_df[target]

print("Features:", X.columns)
```

```
Features: Index(['Year', 'Gas production(m³)', 'Coal production(Ton)',
       'Coal consumption(Ton)', 'Oil production(m³)', 'Oil consumption(m³)',
       'Gas production per capita(m³)', 'Gas consumption per capita(m³)',
       'Coal production per capita(Ton)', 'Coal consumption per capita(Ton)',
       ...
       'Entity_Vanuatu', 'Entity_Venezuela', 'Entity_Vietnam',
       'Entity_Wake Island', 'Entity_West Germany', 'Entity_Western Sahara',
       'Entity_World', 'Entity_Yemen', 'Entity_Zambia', 'Entity_Zimbabwe'],
      dtype='object', length=243)
```

5. Train Test Slpit

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print("Training size:", X_train.shape)
print("Testing size:", X_test.shape)
```

```
Training size: (7389, 243)
Testing size: (1848, 243)
```

6. Model Training

Model Training

```
model = LinearRegression()

model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)
```

7. Model Evaluation Metrics

```python
mse = mean_squared_error(y_test, y_pred)

rmse = np.sqrt(mse)

mae = mean_absolute_error(y_test, y_pred)

r2 = r2_score(y_test, y_pred)

mape = np.mean(np.abs((y_test - y_pred) / np.maximum(y_test,1))) * 100

print("\nRegression Metrics")
print("----------------------")

print("R2 Score:", r2)
print("MSE:", mse)
print("RMSE:", rmse)
print("MAE:", mae)
print("MAPE:", mape)
```

```
Regression Metrics
----------------------
R2 Score: 0.9960781066001961
MSE: 1.4960104489569295e+20
RMSE: 12231150595.740898
MAE: 4585847398.908514
MAPE: 70508810281.43701
```

8. Hyper Parameter Tuning

```python
from sklearn.ensemble import RandomForestRegressor

rf = RandomForestRegressor(
    n_estimators=100,
    random_state=42
)

rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)

print("R2:", r2_score(y_test, y_pred_rf))
print("MSE:", mean_squared_error(y_test, y_pred_rf))
print("MAE:", mean_absolute_error(y_test, y_pred_rf))
```

```
R2: 0.99906730860309
MSE: 3.557761349401742e+19
MAE: 1077952406.5595238
```

9.Actual vs Predicted

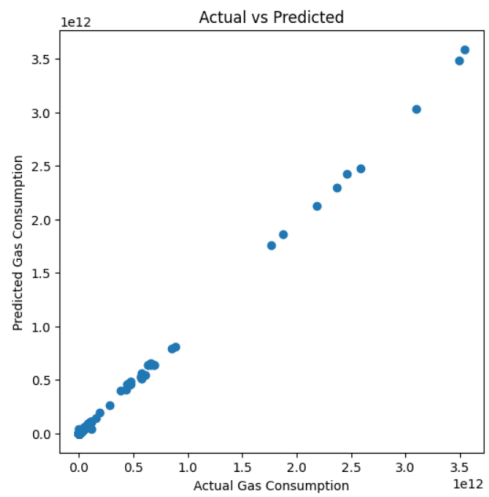
```
plt.figure(figsize=(6,6))

plt.scatter(y_test, y_pred_rf)

plt.xlabel("Actual Gas Consumption")
plt.ylabel("Predicted Gas Consumption")

plt.title("Actual vs Predicted")

plt.show()
```



10. Prediction

```
sample = X_test.iloc[0:1]

prediction = rf.predict(sample)

print("\nExample Prediction:", format(prediction[0], ",.0f"), "m³")
```

```
Example Prediction: 7,061,400,000 m³
```